SIMPLE RAG PROJECT

60 deep workflow explanations with visual diagrams

Each topic page contains a workflow diagram, a deeper explanation of the sequence, and implementation verification guidance. The diagrams describe the required workflow conceptually; repository-specific components must be confirmed from source code.

Page count: 61 total (1 cover + 60 topic pages)

TOPIC 1 OF 60

Project purpose

Workflow diagram



Deep explanation

The workflow for project purpose has a clear handoff between stages. It starts with Goal and moves through the intermediate stages until grounded answer. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Define what the RAG project is expected to do and what problem it solves.

Verification workflow

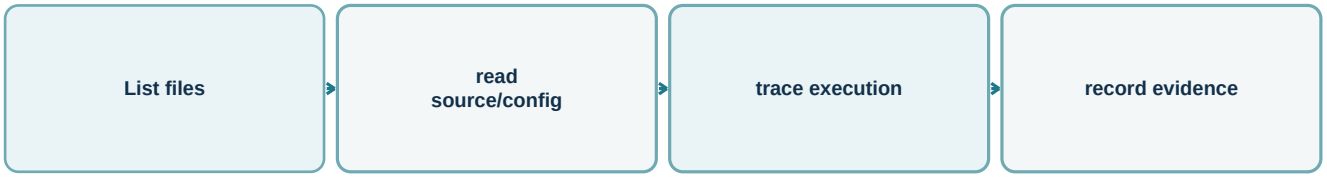
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 2 OF 60

Repository inspection

Workflow diagram



Deep explanation

The workflow for repository inspection has a clear handoff between stages. It starts with List files and moves through the intermediate stages until record evidence. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Inspect the complete repository before writing implementation documentation.

Verification workflow

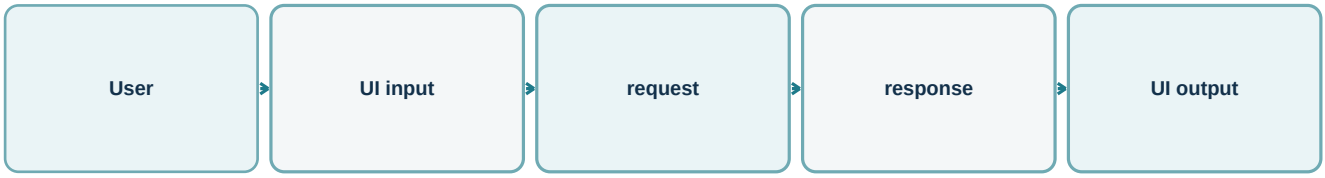
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 3 OF 60

Frontend identification

Workflow diagram



Deep explanation

The workflow for frontend identification has a clear handoff between stages. It starts with User and moves through the intermediate stages until UI output. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Find the user-facing application and explain how it collects and displays questions and answers.

Verification workflow

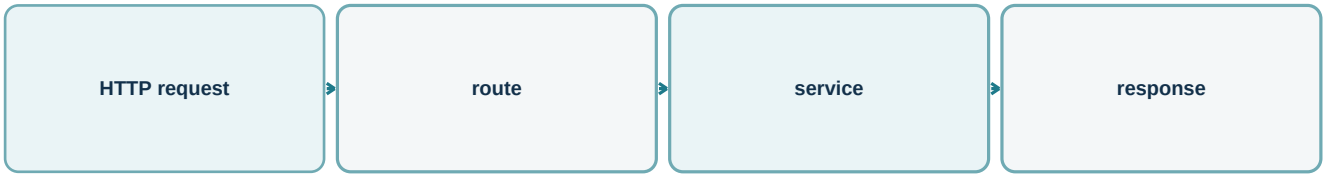
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 4 OF 60

Backend identification

Workflow diagram



Deep explanation

The workflow for backend identification has a clear handoff between stages. It starts with HTTP request and moves through the intermediate stages until response. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Find the server-side application that handles ingestion, retrieval, and generation.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 5 OF 60

Languages and frameworks

Workflow diagram



Deep explanation

The workflow for languages and frameworks has a clear handoff between stages. It starts with Manifest files and moves through the intermediate stages until runtime components. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Inventory programming languages, frameworks, and major libraries.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 6 OF 60

LLM identification

Workflow diagram



Deep explanation

The workflow for llm identification has a clear handoff between stages. It starts with Prompt + context and moves through the intermediate stages until answer. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Identify the exact language model used to generate answers.

Verification workflow

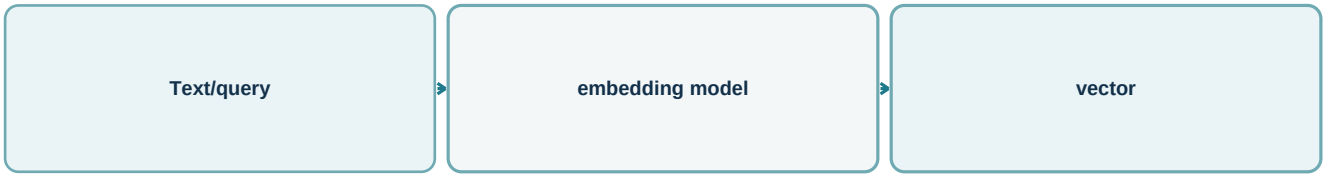
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 7 OF 60

Embedding model identification

Workflow diagram



Deep explanation

The workflow for embedding model identification has a clear handoff between stages. It starts with Text/query and moves through the intermediate stages until vector. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Identify the model that converts text and queries into vectors.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 8 OF 60

Vector database identification

Workflow diagram



Deep explanation

The workflow for vector database identification has a clear handoff between stages. It starts with Chunk + vector + metadata and moves through the intermediate stages until search. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Identify the vector store and explain its role in retrieval.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 9 OF 60

Normal database identification

Workflow diagram



Deep explanation

The workflow for normal database identification has a clear handoff between stages. It starts with Application event and moves through the intermediate stages until result. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check whether a conventional database is used for users, metadata, jobs, or application state.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 10 OF 60

Document source inventory

Workflow diagram



Deep explanation

The workflow for document source inventory has a clear handoff between stages. It starts with Source and moves through the intermediate stages until indexing. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. List exactly where knowledge comes from: files, URLs, records, uploads, or other sources.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 11 OF 60

Document loaders

Workflow diagram



Deep explanation

The workflow for document loaders has a clear handoff between stages. It starts with File/source and moves through the intermediate stages until document objects. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain the loader used for each supported document type.

Verification workflow

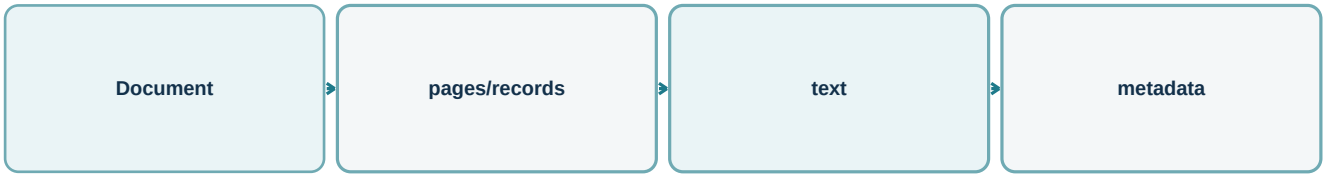
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 12 OF 60

Text extraction

Workflow diagram



Deep explanation

The workflow for text extraction has a clear handoff between stages. It starts with Document and moves through the intermediate stages until metadata. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Show how a raw document becomes usable text.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 13 OF 60

Metadata preservation

Workflow diagram



Deep explanation

The workflow for metadata preservation has a clear handoff between stages. It starts with Source metadata and moves through the intermediate stages until filters/citations/debugging. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain which metadata travels with each chunk and why it matters.

Verification workflow

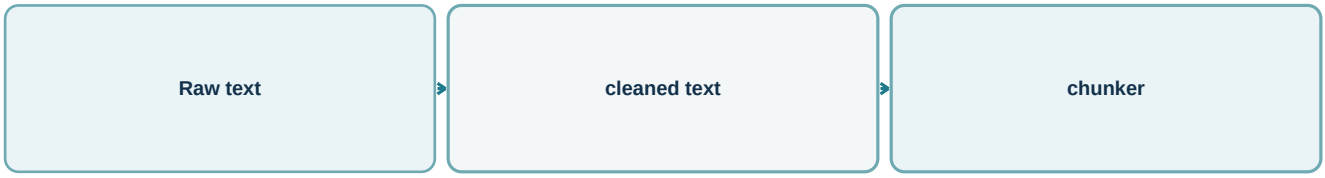
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TOPIC 14 OF 60

Cleaning and normalization

Workflow diagram



Deep explanation

The workflow for cleaning and normalization has a clear handoff between stages. It starts with Raw text and moves through the intermediate stages until chunker. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Identify whitespace cleanup, encoding fixes, or text normalization.

Verification workflow

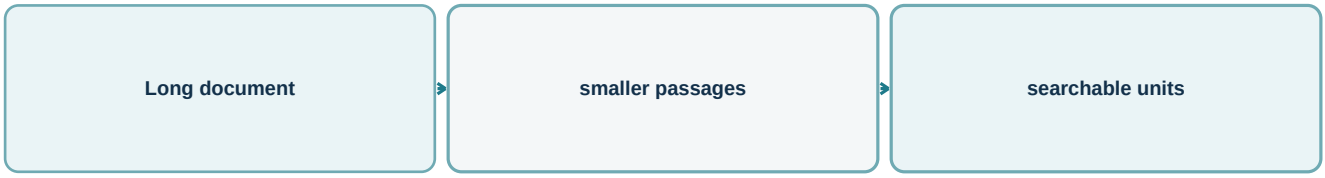
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TOPIC 15 OF 60

Chunk concept

Workflow diagram



Deep explanation

The workflow for chunk concept has a clear handoff between stages. It starts with Long document and moves through the intermediate stages until searchable units. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain a chunk as a small searchable piece of a document.

Verification workflow

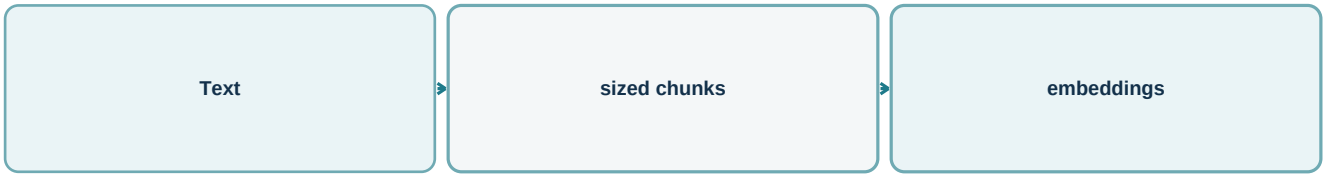
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 16 OF 60

Chunk size

Workflow diagram



Deep explanation

The workflow for chunk size has a clear handoff between stages. It starts with Text and moves through the intermediate stages until embeddings. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Record the configured maximum chunk size and its unit.

Verification workflow

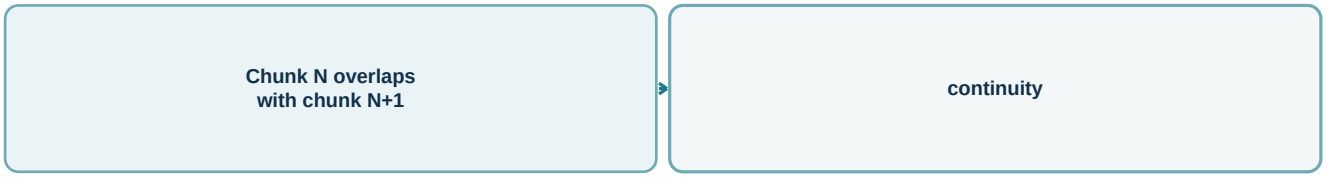
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 17 OF 60

Chunk overlap

Workflow diagram



Deep explanation

The workflow for chunk overlap has a clear handoff between stages. It starts with Chunk N overlaps with chunk N+1 and moves through the intermediate stages until continuity. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Record how much neighboring chunks repeat.

Verification workflow

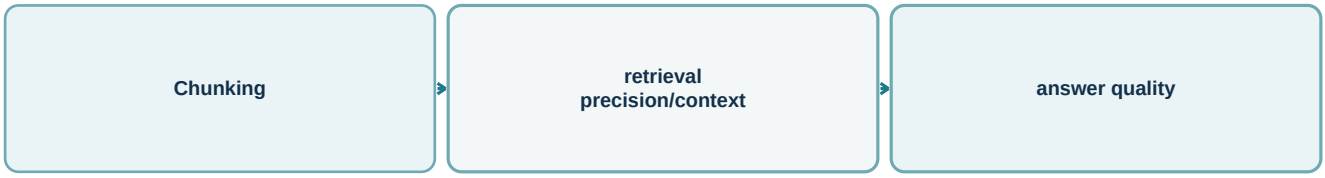
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 18 OF 60

Chunk quality trade-offs

Workflow diagram



Deep explanation

The workflow for chunk quality trade-offs has a clear handoff between stages. It starts with Chunking and moves through the intermediate stages until answer quality. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how overly large or tiny chunks affect retrieval and generation.

Verification workflow

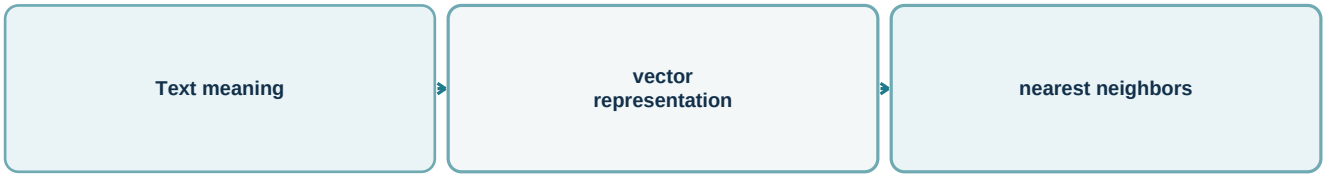
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 19 OF 60

Embedding fundamentals

Workflow diagram



Deep explanation

The workflow for embedding fundamentals has a clear handoff between stages. It starts with Text meaning and moves through the intermediate stages until nearest neighbors. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain why text is converted into numbers for semantic search.

Verification workflow

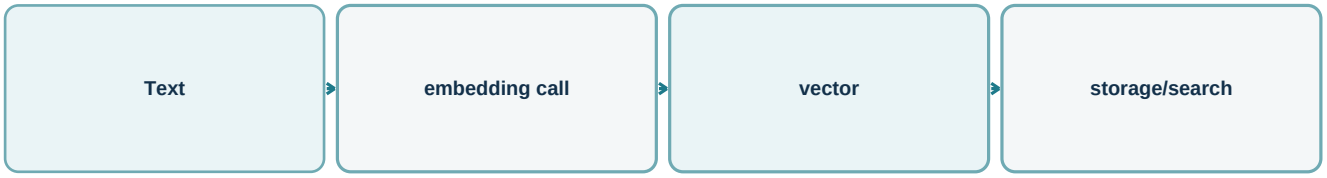
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 20 OF 60

Embedding input/output

Workflow diagram



Deep explanation

The workflow for embedding input/output has a clear handoff between stages. It starts with Text and moves through the intermediate stages until storage/search. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Document the exact input text and vector output path.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 21 OF 60

Embedding dimensions and cost

Workflow diagram



Deep explanation

The workflow for embedding dimensions and cost has a clear handoff between stages. It starts with Model choice and moves through the intermediate stages until database settings. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Report vector dimension and operational cost only when confirmed.

Verification workflow

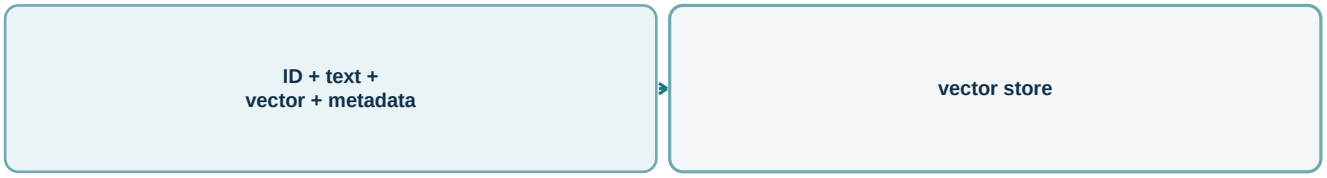
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 22 OF 60

Vector storage schema

Workflow diagram



Deep explanation

The workflow for vector storage schema has a clear handoff between stages. It starts with ID + text + vector + metadata and moves through the intermediate stages until vector store. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Describe how each indexed item is represented.

Verification workflow

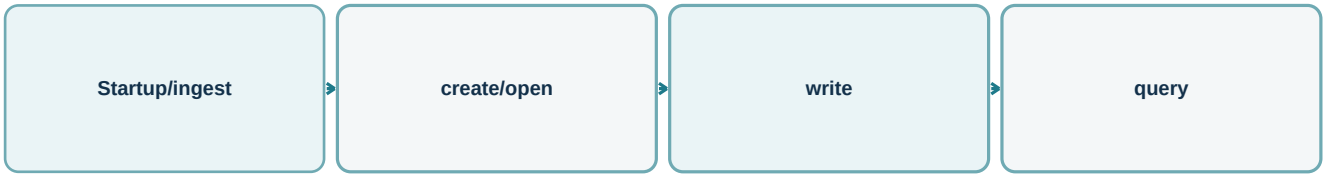
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 23 OF 60

Collection/index lifecycle

Workflow diagram



Deep explanation

The workflow for collection/index lifecycle has a clear handoff between stages. It starts with Startup/ingest and moves through the intermediate stages until query. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain creation, loading, reuse, update, and deletion behavior.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 24 OF 60

Persistence

Workflow diagram



Deep explanation

The workflow for persistence has a clear handoff between stages. It starts with Store and moves through the intermediate stages until data available or missing. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain whether vectors survive process restarts.

Verification workflow

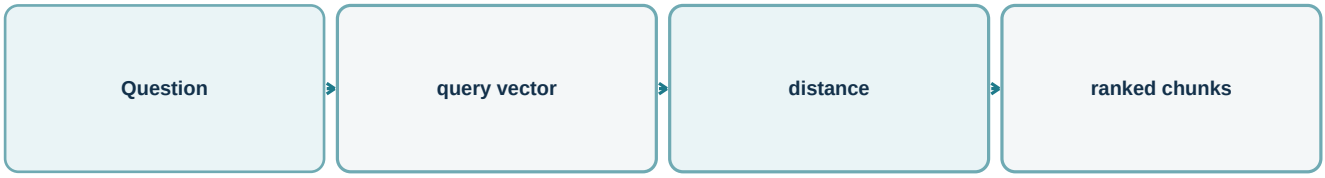
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 25 OF 60

Similarity search

Workflow diagram



Deep explanation

The workflow for similarity search has a clear handoff between stages. It starts with Question and moves through the intermediate stages until ranked chunks. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how a query vector is compared with stored vectors.

Verification workflow

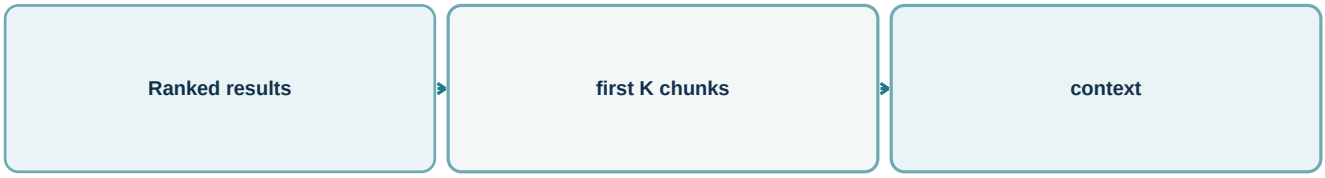
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 26 OF 60

Top-K retrieval

Workflow diagram



Deep explanation

The workflow for top-k retrieval has a clear handoff between stages. It starts with Ranked results and moves through the intermediate stages until context. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain K as the number of candidate chunks returned.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 27 OF 60

Retriever role

Workflow diagram



Deep explanation

The workflow for retriever role has a clear handoff between stages. It starts with Retriever settings and moves through the intermediate stages until documents. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Distinguish the vector database from the retriever abstraction.

Verification workflow

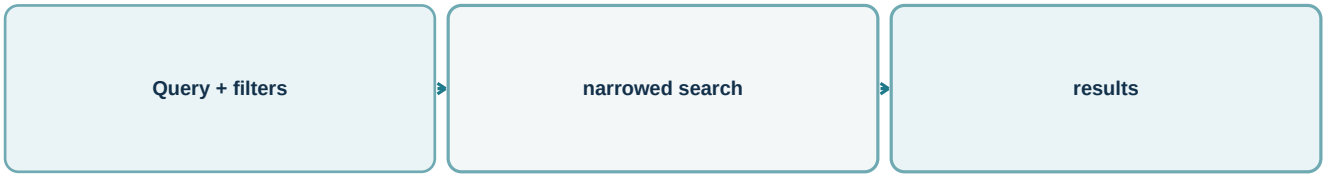
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 28 OF 60

Retriever filters

Workflow diagram



Deep explanation

The workflow for retriever filters has a clear handoff between stages. It starts with Query + filters and moves through the intermediate stages until results. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check for metadata filters, namespaces, tenant filters, or source restrictions.

Verification workflow

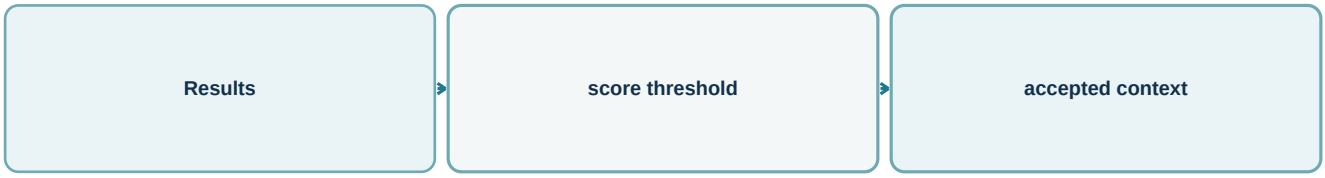
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Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 29 OF 60

Similarity thresholds

Workflow diagram



Deep explanation

The workflow for similarity thresholds has a clear handoff between stages. It starts with Results and moves through the intermediate stages until accepted context. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check whether weak matches are rejected before prompting the LLM.

Verification workflow

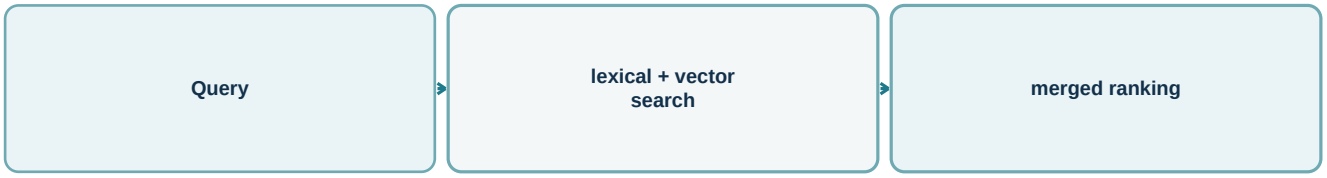
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 30 OF 60

Hybrid search

Workflow diagram



Deep explanation

The workflow for hybrid search has a clear handoff between stages. It starts with Query and moves through the intermediate stages until merged ranking. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check whether keyword and vector search are combined.

Verification workflow

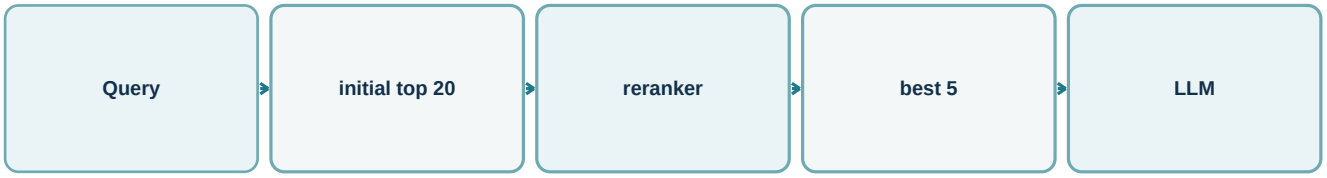
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 31 OF 60

Reranking

Workflow diagram



Deep explanation

The workflow for reranking has a clear handoff between stages. It starts with Query and moves through the intermediate stages until LLM. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain the optional second ranking stage.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 32 OF 60

Query intake

Workflow diagram



Deep explanation

The workflow for query intake has a clear handoff between stages. It starts with Request body and moves through the intermediate stages until query variable. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Trace how a user question enters the system.

Verification workflow

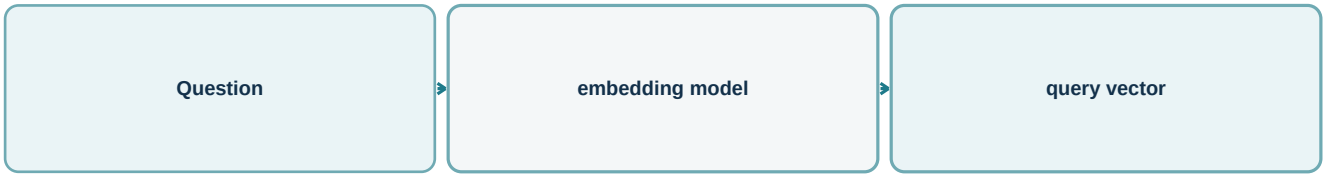
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 33 OF 60

Query embedding

Workflow diagram



Deep explanation

The workflow for query embedding has a clear handoff between stages. It starts with Question and moves through the intermediate stages until query vector. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how the question is embedded for retrieval.

Verification workflow

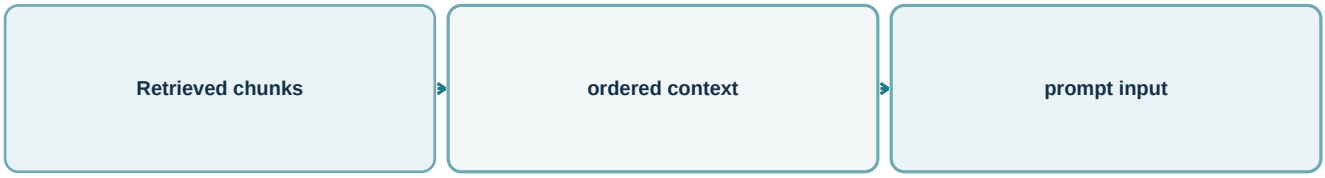
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 34 OF 60

Context assembly

Workflow diagram



Deep explanation

The workflow for context assembly has a clear handoff between stages. It starts with Retrieved chunks and moves through the intermediate stages until prompt input. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how retrieved chunks are combined for the prompt.

Verification workflow

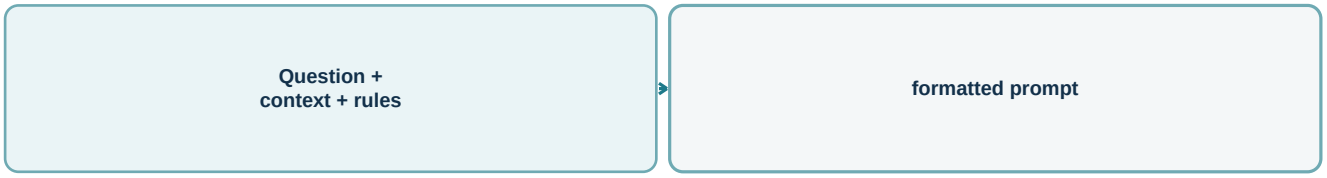
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 35 OF 60

Prompt template

Workflow diagram



Deep explanation

The workflow for prompt template has a clear handoff between stages. It starts with Question + context + rules and moves through the intermediate stages until formatted prompt. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Document the exact instruction template used around the context.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 36 OF 60

Prompt safety

Workflow diagram



Deep explanation

The workflow for prompt safety has a clear handoff between stages. It starts with Untrusted text and moves through the intermediate stages until safer generation. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check for instructions that reduce prompt injection or unsupported claims.

Verification workflow

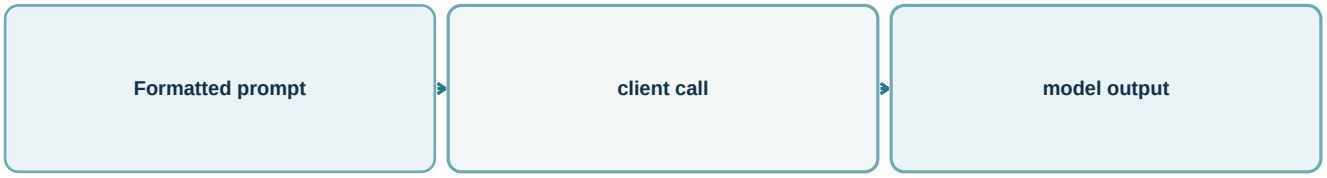
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 37 OF 60

LLM call

Workflow diagram



Deep explanation

The workflow for llm call has a clear handoff between stages. It starts with Formatted prompt and moves through the intermediate stages until model output. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain the code path that sends the prompt to the model.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 38 OF 60

Answer generation

Workflow diagram



Deep explanation

The workflow for answer generation has a clear handoff between stages. It starts with Grounded prompt and moves through the intermediate stages until answer. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how the model turns context into a response.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 39 OF 60

RAG chain

Workflow diagram



Deep explanation

The workflow for rag chain has a clear handoff between stages. It starts with Question and moves through the intermediate stages until return. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Describe how retrieval, prompt construction, and generation connect.

Verification workflow

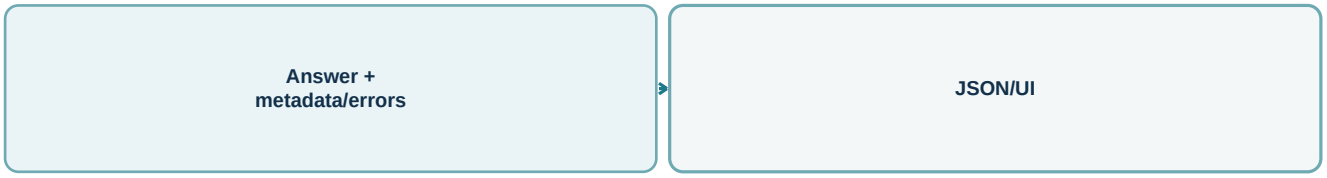
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 40 OF 60

Response format

Workflow diagram



Deep explanation

The workflow for response format has a clear handoff between stages. It starts with Answer + metadata/errors and moves through the intermediate stages until JSON/UI. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Document the API or UI response structure.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 41 OF 60

API endpoints

Workflow diagram



Deep explanation

The workflow for api endpoints has a clear handoff between stages. It starts with Client and moves through the intermediate stages until response. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Inventory routes used for upload, indexing, querying, health, or administration.

Verification workflow

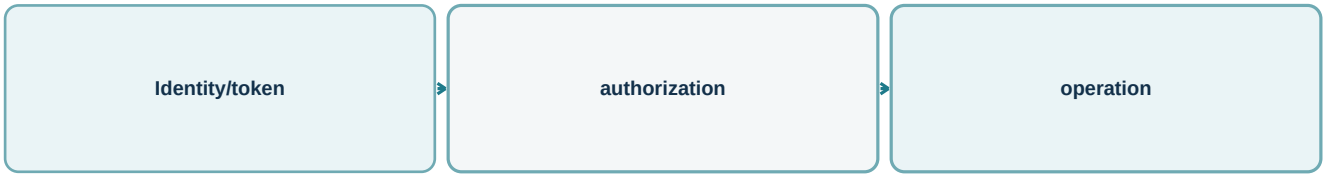
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 42 OF 60

Authentication

Workflow diagram



Deep explanation

The workflow for authentication has a clear handoff between stages. It starts with Identity/token and moves through the intermediate stages until operation. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check whether the project protects APIs or documents.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 43 OF 60

Environment variables

Workflow diagram



Deep explanation

The workflow for environment variables has a clear handoff between stages. It starts with Environment/config and moves through the intermediate stages until runtime behavior. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. List required secrets, model names, URLs, paths, and tuning values.

Verification workflow

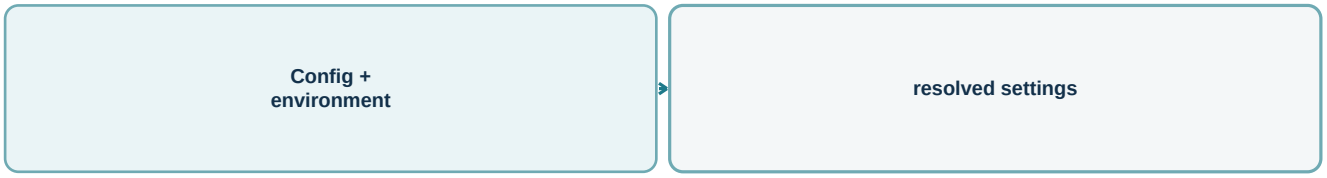
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 44 OF 60

Configuration files

Workflow diagram



Deep explanation

The workflow for configuration files has a clear handoff between stages. It starts with Config + environment and moves through the intermediate stages until resolved settings. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain where settings live and which values override others.

Verification workflow

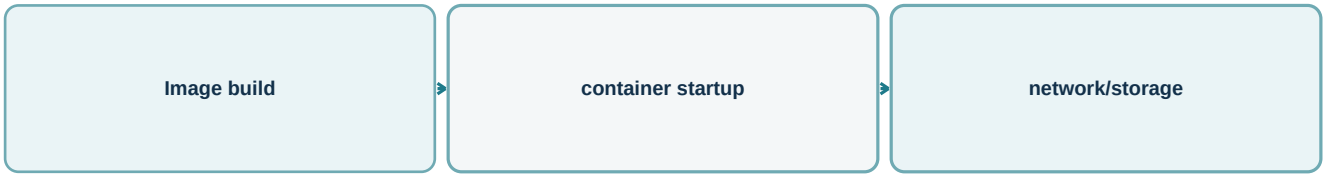
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 45 OF 60

Docker configuration

Workflow diagram



Deep explanation

The workflow for docker configuration has a clear handoff between stages. It starts with Image build and moves through the intermediate stages until network/storage. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check for Dockerfiles, compose files, volumes, ports, and services.

Verification workflow

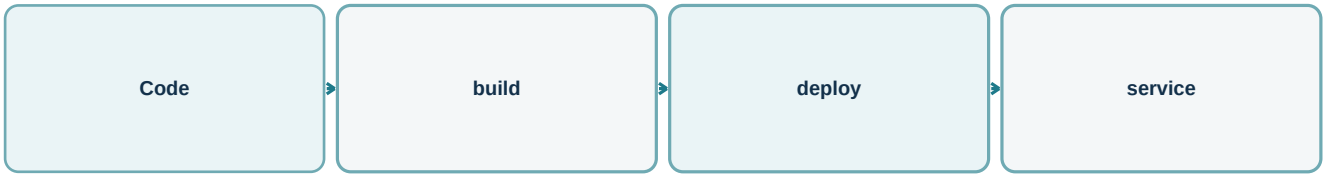
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 46 OF 60

Deployment configuration

Workflow diagram



Deep explanation

The workflow for deployment configuration has a clear handoff between stages. It starts with Code and moves through the intermediate stages until service. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Check for hosting, process, reverse proxy, or cloud deployment instructions.

Verification workflow

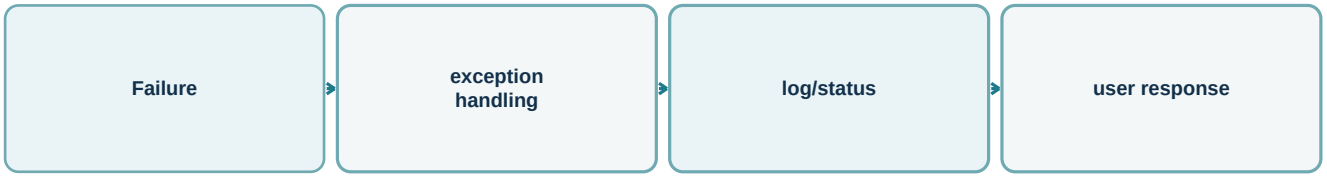
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 47 OF 60

Error handling

Workflow diagram



Deep explanation

The workflow for error handling has a clear handoff between stages. It starts with Failure and moves through the intermediate stages until user response. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Trace failures in loading, embedding, retrieval, and generation.

Verification workflow

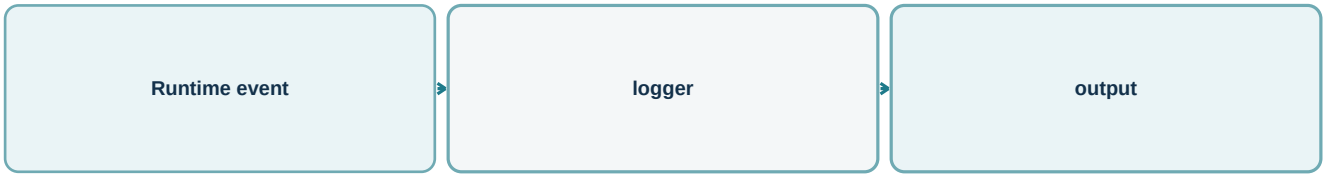
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 48 OF 60

Logging

Workflow diagram



Deep explanation

The workflow for logging has a clear handoff between stages. It starts with Runtime event and moves through the intermediate stages until output. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Identify what events and diagnostics are recorded.

Verification workflow

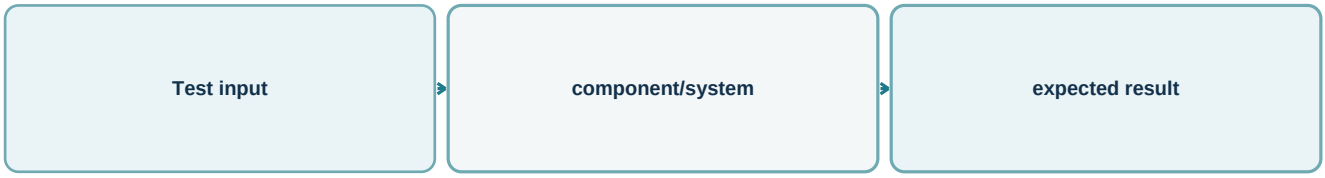
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 49 OF 60

Testing strategy

Workflow diagram



Deep explanation

The workflow for testing strategy has a clear handoff between stages. It starts with Test input and moves through the intermediate stages until expected result. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Inventory unit, integration, API, retrieval, and end-to-end tests.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 50 OF 60

Ingestion workflow

Workflow diagram



Deep explanation

The workflow for ingestion workflow has a clear handoff between stages. It starts with Documents and moves through the intermediate stages until store. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Summarize the offline or startup indexing process.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 51 OF 60

Question-answer workflow

Workflow diagram



Deep explanation

The workflow for question-answer workflow has a clear handoff between stages. It starts with Question and moves through the intermediate stages until response. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Summarize the online query path.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 52 OF 60

Normal LLM vs RAG

Workflow diagram



Deep explanation

The workflow for normal llm vs rag has a clear handoff between stages. It starts with Direct: user and moves through the intermediate stages until LLM. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Compare direct generation with context-grounded generation.

Verification workflow

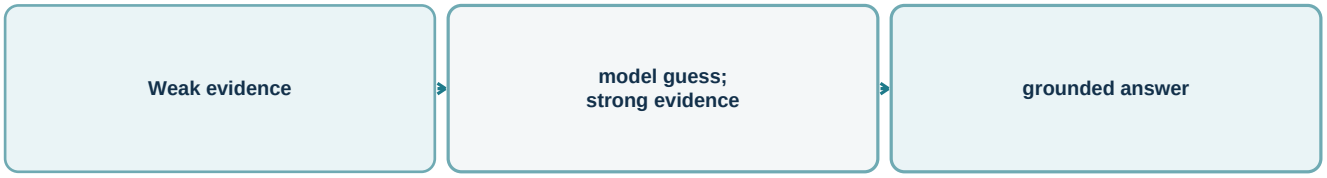
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 53 OF 60

Hallucination and grounding

Workflow diagram



Deep explanation

The workflow for hallucination and grounding has a clear handoff between stages. It starts with Weak evidence and moves through the intermediate stages until grounded answer. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain how unsupported answers can occur and how evidence helps.

Verification workflow

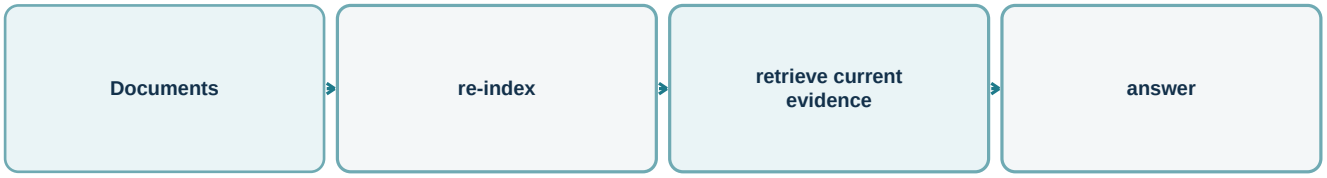
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 54 OF 60

Private and changing data

Workflow diagram



Deep explanation

The workflow for private and changing data has a clear handoff between stages. It starts with Documents and moves through the intermediate stages until answer. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain why RAG is useful for internal or frequently updated knowledge.

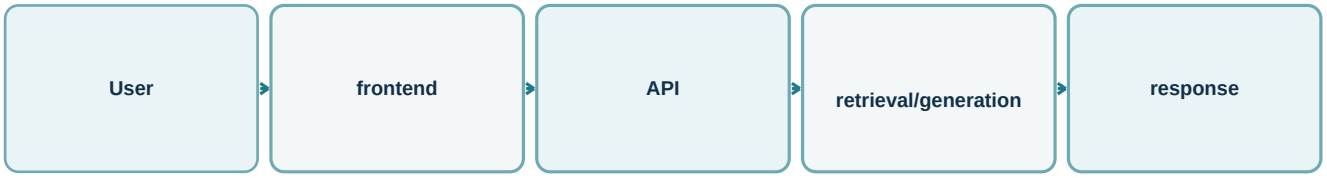
Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

Architecture diagram

Workflow diagram



Deep explanation

The workflow for architecture diagram has a clear handoff between stages. It starts with User and moves through the intermediate stages until response. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Create a diagram that matches actual components and data flow.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 56 OF 60

Beginner explanation

Workflow diagram



Deep explanation

The workflow for beginner explanation has a clear handoff between stages. It starts with Plain concept and moves through the intermediate stages until workflow. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain the project without assuming knowledge of embeddings or vector stores.

Verification workflow

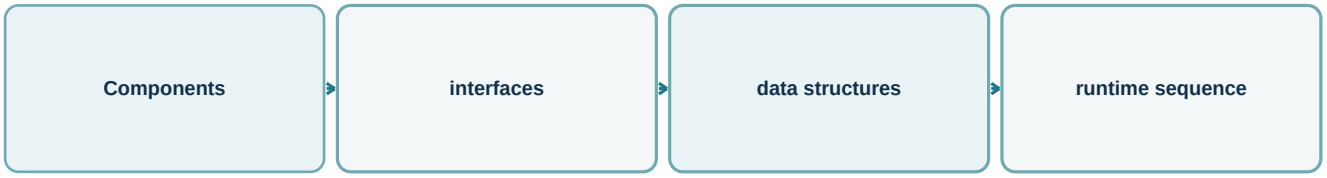
Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 57 OF 60

Technical explanation

Workflow diagram



Deep explanation

The workflow for technical explanation has a clear handoff between stages. It starts with Components and moves through the intermediate stages until runtime sequence. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Explain the same system with precise engineering vocabulary.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

Interview explanation

Workflow diagram



Deep explanation

The workflow for interview explanation has a clear handoff between stages. It starts with Problem and moves through the intermediate stages until trade-offs. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Prepare a concise 1-2 minute explanation of the project.

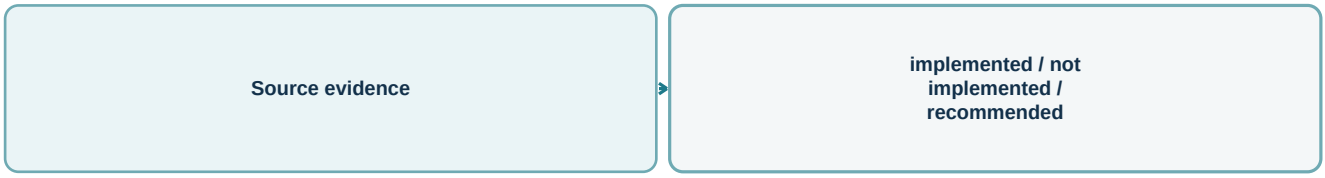
Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

Evidence and gaps

Workflow diagram



Deep explanation

The workflow for evidence and gaps has a clear handoff between stages. It starts with Source evidence and moves through the intermediate stages until implemented / not implemented / recommended. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Separate confirmed behavior, unknowns, and recommendations.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.

TOPIC 60 OF 60

Production readiness review

Workflow diagram



Deep explanation

The workflow for production readiness review has a clear handoff between stages. It starts with Evidence and moves through the intermediate stages until prioritized improvements. The important engineering question is what data is created at each handoff, which component owns it, and what happens when that stage fails. Turn the requirements into a final gap analysis.

Verification workflow

Locate the relevant source files, configuration, dependency, and tests. Trace one real execution path from input to output. Record exact names and values; if evidence is absent, use the required NOT IMPLEMENTED IN THIS PROJECT or PRODUCTION RECOMMENDATION - NOT CURRENTLY IMPLEMENTED label.

Beginner takeaway: follow the arrows from left to right. Each box transforms information or makes a decision before handing the result to the next box.